Augmented Reality and Visualisation of Metadata in the Landscape of Industry 4.0: A Succinct Literature Review

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Abstract.
This literature review explores the use of augmented reality (AR) and metadata visualisation in the context of Industry 4.0, with an emphasis on the integration of IoT elements. Industry 4.0 pursues the digital transformation of business processes and systems to increase competitiveness. However, new studies suggest that AR and mixed reality (MR) can play a key role in this context. This paper analyses publications from 2020 to 2024 in the Web of Science database. Using a filter, 60 publications were selected and then 9 of them were analysed in detail based on refining criteria. The findings suggest a decreasing trend in the number of publications, which may indicate a need for further research and focus in this area. We focused, among other things, on identifying the AR hardware used, where HoloLens 2 is predominant. AR with IoT opens new possibilities in data interpretation and integration, which can be used in rapid prototyping or in the field of manufacturing error prevention. Several studies have revealed significant improvements in machine fault detection and more efficient work for technicians. AR applications are mainly focused on predictive maintenance, manufacturing, and safety.

Keywords: Augmented Reality, Metadata, Visualisation, Information, Internet of Things, Industry 4.0

1 Introduction

The aim of Industry 4.0 is to carry out a digital transformation of business processes, services and systems with a primary focus on increasing competitiveness in day-to-day operations [1]. It is essential to consider innovation within the expanding context of Industry 5.0 and Society 5.0, leading to the concept of Innovation 5.0, emphasising resilience, human-centredness, sustainability, and digitalisation [2]. While implementing technologies such as the Internet of Things (IoT), Cloud Computing, and Artificial Intelligence (AI) into industrial environments is a common practice, extended reality (XR) technologies, particularly AR and MR, are often overlooked. It is important to note that industrial augmented reality (IAR) is a growing area of research and has the potential to be a breakthrough technology [3] [4]. The term XR is referred to by several studies and publications. It encompasses the various forms of reality that exist between reality and virtuality, such as Augmented Reality, Virtual Reality and Mixed Reality [5]. It is important to note that the scientific community inconsistently uses terminology in the field of XR, which can lead to distortions. It is crucial to acknowledge the differences between AR and VR, and to use the appropriate terminology when referring to each technology. The term XR is often used as a generic term that includes both AR and VR, despite their distinct characteristics and continuums [6].

Rauschnabel at al. [6] argue that when the physical environment is integrated into the user experience, it is a form of AR; otherwise it is VR. Exploring deeper into the AR layer, we encounter the concept of Local Presence. This concept is either a form of Assisted Reality with a low level of integration or a form of MR with a high level of integration. Mixed reality is the integration of virtual and real objects, where the two realities (real and virtual) merge and become almost indistinguishable to the user in their extreme versions. However, different authors interpret mixed reality as either synonymous with AR, as a combination of AR and VR, or as a "more powerful" version of AR.

Within Industry 4.0, we can consider autonomous operation of machines that are able to independently collect and analyse data, self-regulate without the need for human intervention, and introduce elements of self-maintenance, self-optimisation, self-recognition, self-adaptation, and other applications of artificial intelligence. This paradigmatic transformation represents a fundamental shift in the understanding and implementation of industrial systems and their ability to make autonomous decisions and adapt to changing environmental conditions [7].

From a technological perspective, the synthesis of various aspects of Industry 4.0 brings together key elements including artificial intelligence, mixed reality and sensors, software combining servitization and systems integration, and cyber-physical systems (CPS) linking computational, networking, and physical processes, especially for smart machines and storage devices. This complex structure has the potential to provide technological innovation, which is the result of applying technology to improve transparency, scalability, responsiveness and flexibility in the supply chain and to enable real-time information gathering [8] [9].

IoT then refers to the networking of devices within the Internet or local area network, which includes computing units, devices, and sensors. In an industrial setting, the term Industrial Internet of Things (IIoT) is commonly used to characterise its IoT iteration, notwithstanding the frequent interchangeability of these terms within the industrial domain. The integration of smart devices, advanced monitoring systems, and digital information processing through IoT and the Cloud is fundamentally transforming the approach to solving security challenges in various industrial sectors. Within industrial applications, IoT is associated with cloud-based solutions that collect data from various sources such as sensors and machinery. Current advances in communication standards, exemplified by Wi-Fi 6, seek to increase data throughput, improve energy efficiency and optimise the judicious use of spectral resources [10] [11] [12]. In parallel, the era of mobile networks, referred to as 5G, is expanding the horizons of the IoT by significantly increasing data transfer speeds and correspondingly reducing latency [13].

Although information and communication technologies (ICT) undoubtedly play a key role in this process, it is important to note that the human factor is equally important. The technical aspects of smart infrastructure architecture in a smart environment include physical elements such as devices and sensors, as well as software structures such as data, analytics, feedback, and adaptability. The impact of these structures on the interaction between humans and technical devices highlights the relationship between humans and machines [14] [15]. Perri et al. [16] provide valuable insights aimed at transforming user behaviour in manufacturing. This issue is particularly important in the context of the deployment of smart networks and intelligent machines. Data is an essential element for connecting the physical and cyber worlds. They are crucial for the interaction of digital twins, which serve to represent physical entities and at the same time enable the extraction of knowledge [17]. Similarly, Mofolasayo et al. [18] describe this aspect as an expected result of integrating Industry 4.0 with lean manufacturing principles - achieving "information transparency". This transparency consists in connecting data and information during the production process to optimise the process itself. Among other key areas, the use of blockchain can be included, which is essential for the development of applications in areas such as security, networking, computing, resource management [19].

Introduction of Augmented Reality in Industry 4.0: Shaping Data and Visualising Insights

Augmented reality sits on a continuum between real and virtual environments, while being integrated into the more general area of mixed reality [20]. Azuma defines augmented reality as a system that demonstrates three key characteristics: it combines real and virtual elements, enables real-time interactivity, and is recorded in three dimensions. The concept of augmented reality can be seen as a different approach to technology that avoids a complete replacement of the real world. Augmented reality seeks a mid-way point between a completely synthetic and a totally real environment in order to achieve the effect of the coexistence of these two worlds [21].

The effective integration of extended systems is a key element in the context of the digital transformation in manufacturing, considering the concept of Industry 4.0. Despite the positive findings of scientific studies to date, which highlight the significant potential of XR applications in manufacturing environments, there are limitations in the actual use of XR systems by engineers in their regular working practices [22]. AR is currently most developed in maintenance, followed by assembly, human-robot collaboration, manufacturing, training and logistics [23]. Although there are several positive aspects of AR, this technology cannot yet be considered fully mature and suitable for full deployment in an industrial environment [24]. Exploring a form of visualisation for augmented reality, the AnnHoloTator platform based on Hololens 2 was introduced, providing users with the ability to visualise, manipulate and annotate digital documents in 3D space with data-flexible and collaborative features [25]. The ASMARI prototype consistently interprets and visualises key informational aspects of the Human Machine Interface (HMI), allowing users to view statistics and issue basic commands without having to be constrained by a specific console. This innovation significantly increases user situational awareness and overall productivity [26]. AR innovates information visualisation by enabling the superposition of virtual 3D models onto physical prototypes, allowing workers to easily identify manufacturing and assembly errors and then add 3D annotations directly to the virtual models to report and share information with the engineering department [27].

Several experiments in XR are related to user training, remote labs and operations, assembly processes, performance, decision making, spatial mobility, perception and situational awareness, visualisation, and ergonomic design in the context of augmented reality. XR technologies have been positively evaluated in environments where they offer significant opportunities, whether virtual, augmented, or mixed reality. The presence of sensors in applications helps to reduce the differences between these types of XR devices [28].

AR has been successfully used in studies to improve the presentation of information and intuitively demonstrate setups on machine tools in industrial environments focused on the production of various components [29]. The use of augmented reality brings major improvements in the visualisation of elements. The implementation of AR in welding processes has greatly improved the ability of workers to visualise intricate details, leading to improved welding results compared to traditional methods [30]. ARTool Zero is an example developed to assist operators in programming touch probe trajectories, generating customised programs and simulating probe guidance to identify geometric features [31].

Augmented reality can be applied successfully in information-intensive tasks, but in machining the possibilities are limited due to the prevalence of machine processes. AR is gaining importance in industries, especially as a necessary component of the Industry 4.0 framework, indicating a move towards greater competitiveness, innovation and performance excellence [32].

2 Literature Review Methodology

Here we present a brief literature review, generated through a filtered survey, focused on evaluating current technology under the overarching theme of augmented reality and visualisation in industrial environments. The methodological basis of our systematic literature review adheres to established principles and follows a process of problem identification and subsequent literature review [33]. Importantly, this overview is deliberately limited to be concise and offer a focused perspective on the chosen issue. Our approach involves a thorough examination of numerous scientific papers and publications, subjected to peer review by experts from both academic and practitioner backgrounds. The filtering process is based on an explicit definition of the problem to be addressed. Subsequently, a careful selection of relevant publications is made to serve as the basis for generating meaningful insights and results [34]. This method provides a robust assessment of the current state of knowledge and presents a comprehensive understanding of the applications and implications of augmented reality and visualisation technologies in an industrial context.

2.1 Research Procedure Selection

In this research approach, we relied on a methodology that elaborates on the steps necessary to conduct a systematic literature search and process the search outputs. Our systematic review aims for a narrow and specific framework in contrast to traditional search procedures involving a comprehensive review of the published literature. Given the newness of the topic under investigation, the initial stages of the research began with a thorough exploration of this emergent area [35].

We have anchored ourselves in a similar framework that defines a systematic search procedure as a survey of many peer-reviewed publications over several years from which rich data can be extracted for analysis. The methodologically filtered procedure allows us to gather relevant information through a clearly defined approach, which reduces systematic errors. This approach involves the identification, evaluation and synthesis of all relevant studies using the chosen methodology. As part of our narrow systematic literature review, we built on this methodology [36].

First, we focused on research planning, where we identified gaps in existing research and formulated research questions leading to the identification of relevant publications. This process ends with determining a plan and method for resource exploration, which includes selecting appropriate software, limiting criteria for relevant publications, choosing databases to search.

Afterwards, the actual investigation and implementation of the research takes place. The first step was to identify primary sources to serve as the basic starting points for the research. We then selected sources according to the quantity corresponding to the defined keywords in the research question. The keywords were organised into a research question and the software was used to select the quantity of relevant publications.

The identified publications were then analysed according to predefined criteria. After removing irrelevant publications, a qualitative assessment was conducted, which included removing possible duplicate sources and manually checking the publication title, abstract and focus. This procedure ensured that relevant information was obtained and allowed for comparison of individual publications or groups for subsequent evaluation.

The final phase focuses on evaluating the entire research and generating a report. To begin with, all scientific publications are evaluated statistically to obtain basic information about the research conducted up to that time. The last step of the whole systematic review process is to summarise the results from the authors' perspective and answer the research questions set at the beginning of the paper.

The aim of this procedure is to review a narrow range of scientific publications, minimise the inclusion of irrelevant sources and avoid errors in literature searches. Following this procedure, it was possible to record, process and identify relevant papers related to the identified research field. The principle underlying this simplified systematic search is based on the principles of mixed methods research, which combines quantitative and qualitative approaches. The quantitative approach was used to search selected citation databases. For this simplified approach, we focused only on searching the Web of Science Core Collection (WoS) citation database.

2.2 Query Formulation

To create a systematic literature review, it is necessary to analyse the area of challenge based on the thematic discussion that takes place in the opening chapters of the paper. The various steps of this research process not only conceptually summarise the research process, but also provide a thorough description of the problem area, formulate research questions, and then answer these questions analytically. In the introductory sections, the topical relevance of the research topic within Industry 4.0 was emphasised. As a result of this recognition, the research focus naturally shifts to augmented reality, as one of the fundamental keystones of Industry 4.0, and information visualisation in an industrial context.

A definition of the research area is proposed, and the research topic is defined: identification and development of augmented reality in the context of Industry 4.0. Specifically, the aim is to identify the stage of development of information visualisation, and thus metadata, through augmented reality technology.

From this explicit definition, we can formulate specific research questions that will lead to a deeper understanding of the issue under investigation. This stage is a key step in our systematic approach as it allows us to manage the course of the research process in a more structured way. In this way, we create a solid framework for our systematic investigation, which aims not only to gather relevant information but also to provide a deeper understanding of the dynamics and perspectives of augmented reality technology within Industry 4.0.

Now let us formulate the research questions (RQs). All RQs will refer to the specific literature sources found, which will be refined based on the stated requirements.

RQ1: What is the general motivation for research in IoT and AR?
RQ2: What hardware and combinations of hardware are used?
RQ3: What software is used and what are the most common algorithms used for object detection in space?
RQ4: What are the common areas of real-world applications of IoT and AR?
RQ5: What is the proportion of connections to real-time data in the sample studied?

2.3 Search Query Formulation

Keywords were identified for the systematic search based on the research questions. This was done by constructing a search query using the "AND" and "OR" principles. The keywords and the entire search query were derived from the scientific publications cited in the introduction of this paper, which provide the basic framework for the entire research and define the objectives of this research. The scope of the search was limited to publications from 2020 and younger. This limitation was chosen to provide a contemporary view of the trend and status of the research area over the last five years. With this decision, we sought to determine whether interest and research in this area is expanding or contracting, as well as what the current state of research is and what achievements can be identified.

The specification of the search query involved searching the Web of Science database. To implement this search, a filter was used directly within the WoS database interface. This method was used to quantitatively identify relevant publications that are directly related to the search query. For further fine-tuning, the search results were limited to the basic criteria.

Research limitations - Specification of the paper topic based on the occurrence of keywords through the search query (publication title, abstract, keywords) according to the query:

(Internet of Things OR Sensors OR IoT OR IIoT) AND (Visualisation OR Human-centered computing OR Interface) AND (Augmented reality) AND (Data OR Production Data OR Machine Data) AND (Industry OR Industry 4.0 OR Industry 5.0)

The generated search query is shown in **Figure 1**, which directly reflects the filtering performed in the Web of Science database. The relevant information is the total number of publications that match the specified filters within this database. This figure identifies the selection of publications that are associated with each individual filter. Here, substitution is used, and these filters are then combined with other filter layers, further refining the selection specification.

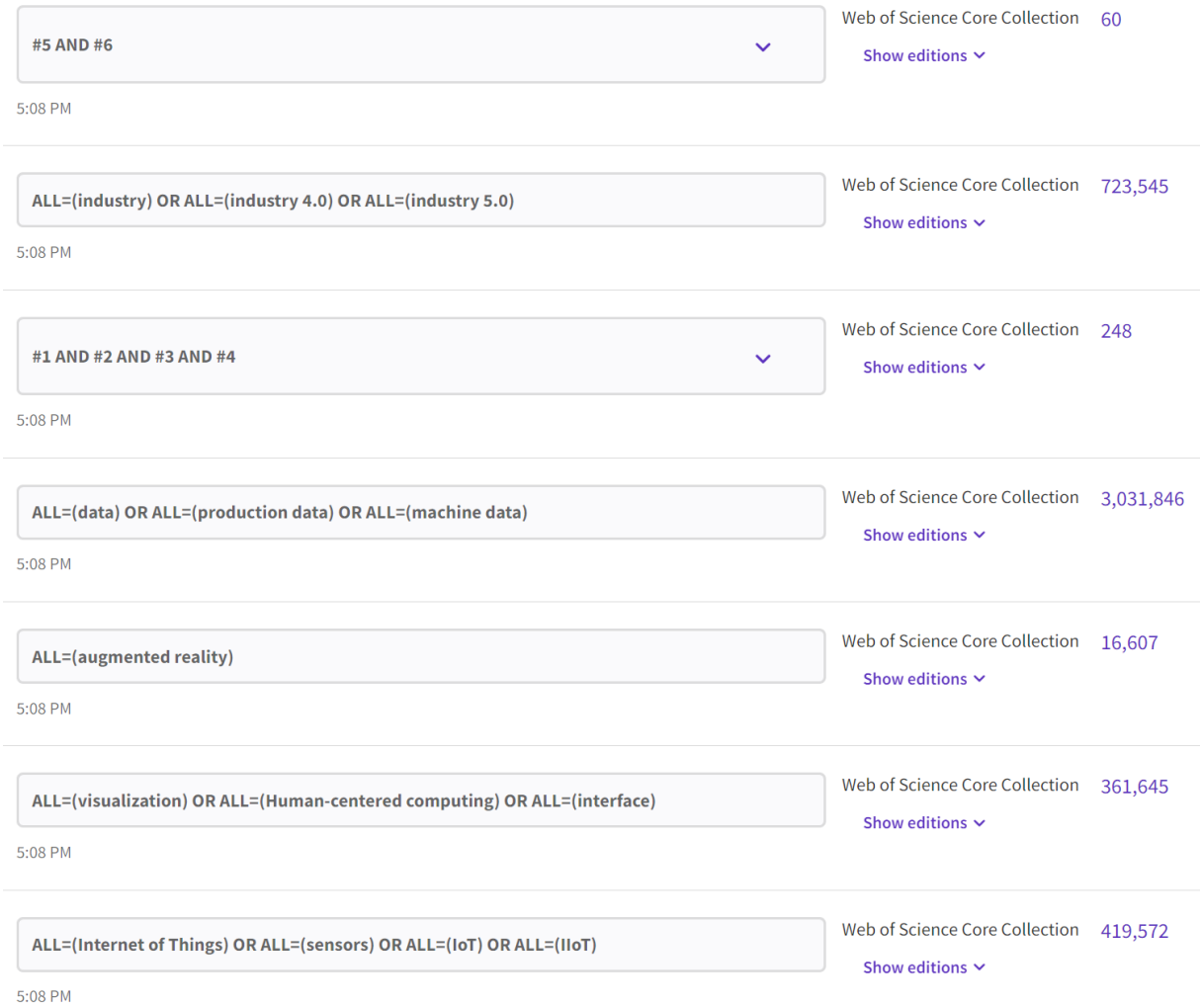

Figure 1 Formulating a search query in the Web of Science database

The research parameters include a requirement that publications be in English, with an emphasis on the period 2020 to 2024 as the years of publication. The types of publications listed include conference papers, conference proceedings, papers, books, book chapters and research papers, representing the target sources for the succinct systematic literature review.

2.4 Research Results

The research was carried out during January and February 2024. A quantitative examination of scientific publications within the selected database based on the established search query yielded exactly 60 relevant publications that fall within the last 5 years. In the next phase of the iterative process, these results were subjected to a rigorous qualitative assessment of relevance. This step involved identifying the origin of the publications and assessing their credibility. Subsequently, the content of the publications was manually checked, with an emphasis on the specific focus of the title and abstract in relation to the issue at hand. As a result of this analysis, only 9 publications were determined to meet the relevance criteria for the next stage of the research. Namely, the following conditions must be met: the study focuses on the engineering or manufacturing industry within the Industry 4.0 concept; the case study can be a laboratory experiment designed with an Industry 4.0 application in mind; the main topic of the paper is the role of augmented reality and its implementation in an industrial environment, with an emphasis on the visualisation of virtual artifacts and metadata that can be presented either in real time or in static form.

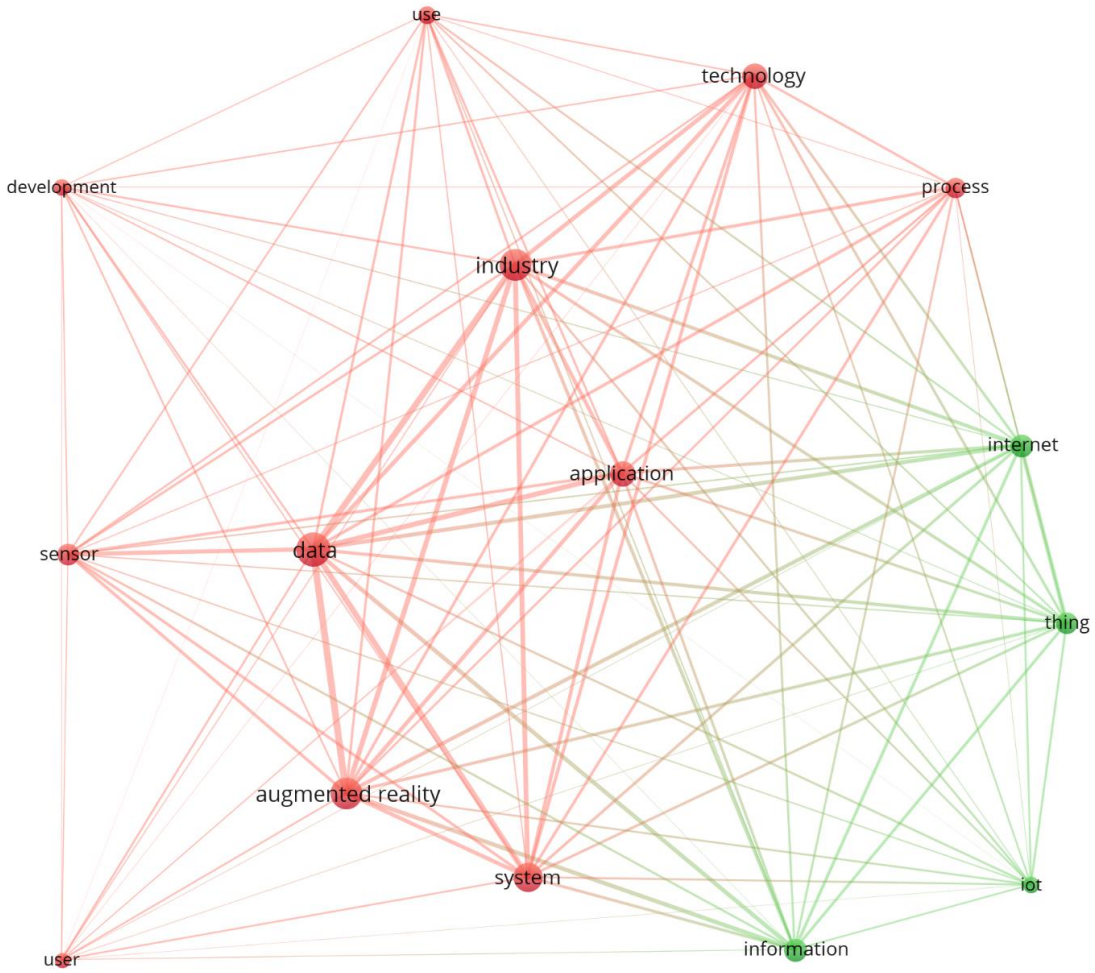

Figure 2 Analytical presentation of the distribution of keywords and their interactions in accordance with the search query using the VOS visualization tool

Using the visualisation tool VOSviewer, we analysed the occurrence of keywords in selected papers from the WoS database, see **Figure 2**. The main connections between these papers were observed in the areas of augmented reality, data, and industry. Important links are found between keywords such as "system" and "application", "technology" and "process". Other keywords such as "user", "sensor", "development", "use", "Internet of Things", "information" and "IoT" are less prominently represented in these papers. This selection of keywords is derived from the analysis of 60 papers that were selected based on a specific filter, as shown in **Figure 1**. This visualisation allows for a more efficient identification and linking of key themes among the selected papers, which contributes to a better understanding of the topic.

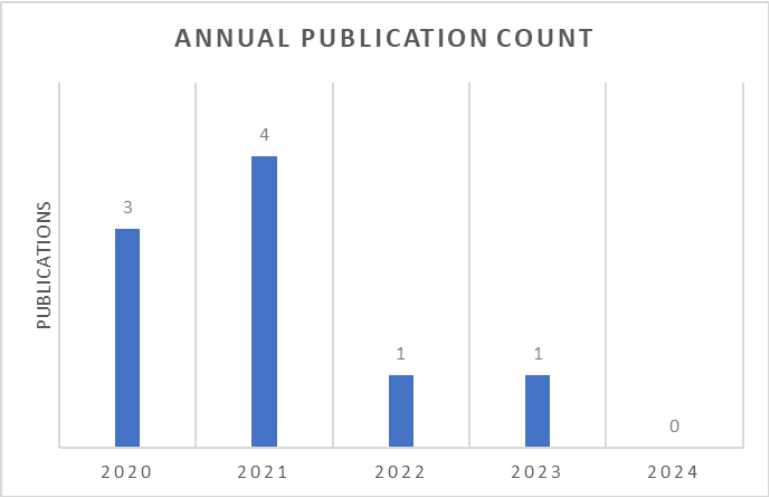

Figure 3 Temporal trends in publications focusing on the integration of visualization and Metadata in Augmented Reality from 2020 to 2024

The analysis of **Figure 3** allows a detailed tracking of the frequency dynamics of publications between 2020 and 2024 that focus on the use of visualisation and metadata in augmented reality with an emphasis on IoT. This timeframe is key to understanding developments and trends in the field in recent years. Based on a narrow selection that considers specific research criteria, we looked for publications that make a significant contribution to the defined topic. The graph shows that the most research activity occurred in 2021, indicating a peak point. However, there is a rapid decline in the frequency of publications thereafter. This downward trend can be interpreted in different ways. One possible explanation is that advances in AR visualisation and metadata have reached a certain saturation, causing a decline in new publications. It can also be interpreted as a blind research path. Another possible factor could be insufficient advances in technology and hardware, which hinder the further development of this field.

Table 1 Shortlisted publications and their authors by number of citations

Authors	Number of Citations	Title of Publication	Year
Angrisani et al.	60	A Wearable Brain-Computer Interface Instrument for Augmented Reality-Based Inspection in Industry 4.0 [37]	2020
Mourtzis et al.	14	Integrated and adaptive AR maintenance and shop-floor rescheduling [38]	2021
Verde et al.	13	Advanced Assistive Maintenance Based on Augmented Reality and 5G Networking [39]	2020
Rosales et al.	8	IIoT based Augmented Reality for Factory Data Collection and Visualization [40]	2021
Lalik and Watorek	7	Predictive Maintenance Neural Control Algorithm for Defect Detection of the Power Plants Rotating Machines Using Augmented Reality Goggles [41]	2021
Kim et al.	7	A Hybrid Approach to Industrial Augmented Reality Using Deep Learning-Based Facility Segmentation and Depth Prediction [42]	2021
Husinsky et al.	3	Situated Visualization of IIoT Data on the Hololens 2 [43]	2022
Stacchio et al.	1	AnnHoloTator: A Mixed Reality Collaborative Platform for Manufacturing Work Instruction Interaction [25]	2023
Gross and Wuest	0	Smart Products in Smart Manufacturing Systems: An Opportunity to Utilize AR? [44]	2020

Table 1 presents a comparison of the number of citations for selected publications with the number of citations by authors and the year of publication. This set of publications, selected based on the narrow selection criteria of the literature review, includes mostly publications with a minority of citations, except for Angrisani et al. [37]. The remaining publications are in the top ten citations with [38] and [39] doing slightly better in terms of citations.

3 Summary of Theoretical Knowledge

The evolving fields of AR and IoT in industrial applications are bringing innovative approaches to optimising manufacturing processes, predictive maintenance and improving worker efficiency. This dynamic combination of technologies opens new possibilities for monitoring, visualising, and controlling industrial operations. In this context, the integration of hardware and software aspects is also a key element. Below, we analyse nine research papers that examine in detail the implementation of AR and IoT in an industrial setting. Each paper brings a unique perspective to the issue and provides valuable insights that we will systematically explore and compare. We will focus on the common features of these papers in correlation with the above research questions, whether in terms of research motivation, hardware application, software used and data linking, object detection, or specific industrial application areas. This analysis will provide us with a close insight into current innovations in AR and IoT from an industrial perspective. We will look at the synergies between these technologies and assess their potential impact on industrial processes through a table delineating key areas.

Research motivation

Augmented Reality support is an effective means of minimising losses caused by poor product quality or process failures. The ability to work with, interpret and connect data within an AR environment, especially with the widespread use of IoT, offers new opportunities. The digitisation of data and processes aims not only to reduce production time, but also to prevent errors and mistakes and to facilitate rapid prototyping and the design of innovative products. Monitoring manufacturing processes and product status using AR is expected to enable employees in the manufacturing environment to proactively prevent and resolve product-related errors [44]. The support to the workers is twofold: new employees should be able to quickly and easily understand the ongoing processes in the company, while maintenance elements that are usually invisible from the outside, e.g., of a production machine, should be displayed in a form that can be understood and easily accessed directly at the equipment using AR [43]. The development of artificial intelligence and deep learning algorithms greatly enhances the concept of predictive and preventive maintenance and monitoring the condition of industrial equipment [41]. The use of AR enables the visualisation of the machine condition, maintenance planning based on the current production schedule and remote maintenance execution [38]. The integration of diverse sensors with augmented reality visualisation could greatly improve the safety of operators in hazardous industrial plants [39].

Hardware application

Augmented reality, combined with IoT, low-cost sensors and affordable programming languages, is moving the industry forward strongly. The implementation of a sensor system based on the Arduino platform enables data transfer to AR devices. A similar approach was recently taken by Husinsky et al. [43], who created a proxy application using Node-RED and an OPC-UA module extended for Raspberry Pi 4. Lalik and Watorek [41] extended the hardware base of integrated AR glasses by connecting a FASTCAM Mini AX50 camera from Photron to increase the frame rate. In a similar vein, [39] integrated an external FLIR AX8 thermal sensor into a software scene recognition system to detect and signal harmful thermal sources. Rosales et al. [40] implemented machine feedback such that each virtual artifact on a smart device represents a signal or sensor, with its colour indicating its real-time state. Likewise, [37] use temperature and humidity sensors to rapidly control machines.

Most scientific publications favour HMD glasses as AR devices, particularly Microsoft's HoloLens 2, or in some cases unspecified AR HMD glasses. The only exceptions to the above papers are the two cases where a mobile phone or tablet is used as the primary AR device.

Software used and data linking

How authors approach the use of software and design architecture varies depending on the specific situation and environment in which the AR application is being developed. When focusing on data integration with the HoloLens AR device, it is essential to create an interface to an industrial network. In some cases, communication using the OPC-UA (Open Platform Communications - Unified Architecture) standard and subsequent translation of the data into MQTT format is applied. Implementing the conversion to the MQTT format using Node-RED allows easy integration of HoloLens 2 with other devices in the IoT environment that also use this communication protocol [43]. For example, Rosales et al. [40] used an OPC server such as KEPseverEX and then transferred the data to the Industrial Internet of Things (IIoT) platform ThingWorx, where the configuration process was automated using the ThingGenerator application.

Many cases use wireless remote access to data that is then displayed in the AR environment. This approach is chosen due to the insufficient processor power of the AR device to process the data locally. All these processes are processed outside the AR device thanks to the cloud approach [41]. For instance, a cloud-based machine monitoring platform can be implemented using a Linux virtual machine IaaS (Infrastructure as a Service) using Apache HyperText Transfer Protocol Secure (HTTPS) server, Ruby on Rails (RoR) framework and MongoDB (NoSQL) database [38]. Other platforms work in a similar way [42] [25].
The development of artificial intelligence and deep learning algorithms strongly supports the concept of predictive and preventive maintenance and condition monitoring of industrial equipment [41]. Kim et al. [42] implemented an intelligent feature where they combine instance segmentation and depth prediction using TensorFlow or Pytorch in WIN10 environment to detect and segment objects from an AR view image. The module of region recognition and extraction uses a mask convolutional neural network (Mask R-CNN) to detect objects and segment them from the RGB image. This can predict the depth and estimate the 3D map. Their approach to software development is unique in this field. Angrisami et al. [37] combined AR with a brain-computer interface (BCI), which allows hands-free control of HMD glasses by monitoring brain activity using electroencephalography (EEG).
One positive aspect of the reviewed publications is that most of them work with real-time data, either directly from machine sensors or from database systems that change dynamically. A few cases also work with static data, but real-time data is being considered for further development [23], [40] and [42].

Object detection
Any placement of virtual content in an augmented reality environment requires a specific object or location. Detection can be divided into two approaches: traditional and innovative.
One of the traditional detection methods is the use of Bluetooth technology. The Android app automatically locates Bluetooth transducers connected to AR glasses. Once they are found, the closest transducers with the strongest signal are displayed in the AR glasses and the user is prompted to select and connect them. After successful connection, the transducer sends the measured data [37]. To link the visualised results to each individual machine, a unique AR tag is placed on the machine [38], [41]. The centralisation of the virtual object can be done using a manual method that allows fast and accurate localisation of the 3D model at the right place. This process is achieved by defining three points in the virtual space from which two vectors are derived and then a coordinate system is created using an orthogonalisation procedure [43]. Another option is using feedback from an Arduino device that is hardwired into the product and actively monitors motion and temperature, providing useful feedback throughout the manufacturing process [44].
Advanced and original technologies include depth sensors, which are used in devices such as HoloLens. These sensors enable the visualisation of obstacles by processing 3D coordinates within Multi-access Edge Computing (MEC) technology. This way, holograms are projected into an augmented reality field based on the spatial position of the detected objects [39]. Spatial anchors are used to anchor the visual content in the virtual environment, which can be created using any augmented reality (AR) enabled devices. These anchors serve as virtual markers that define a physical reference in the real environment where the digital content can be displayed [40]. An alternative approach to the solution is the use of deep learning and neural networks. For this scenario, the AR image from the smart device is sent to a remote server to perform deep learning. This process involves segmenting and predicting the depth of physical objects in the AR image. The results are then transmitted back to the AR client to generate 3D models and solve visual problems [42].

Application area
Most of the publications in this field are based on experiments carried out under laboratory conditions. During these experiments, the connected elements of augmented reality, IoT and data visualisation are tested, developed, and refined. A positive finding is that several of these papers have been implemented in real industrial environments. One such example is the paper by Rosales et al. [40], who study the integration of data from manufacturing environments into the IIoT and its real-time visualisation using augmented reality and mixed reality on smart devices at Volvo Group. Husinsky et al. [43] focused on the design of solutions and process support in industrial furnaces for aluminium smelting, presenting the challenges associated with the use of augmented reality in industrial environments.

The application of these technologies in industry is mainly in the areas of predictive maintenance and manufacturing, which are partially complemented by inspection and safety aspects.

All the findings were systematically summarised and presented in **Table 2**, which ensures clarity and easy orientation. This table serves as a compact and structured overview of all the relevant information, allowing for quick comparison and analysis within a succinct literature search. These citations refer to February 2024.

Table 2 A systematic review of key findings identified in a literature review of key publications

Authors	AR hardware	Real-time data	Industrial area	Hardware	Software	Object detection	Implementation environment
Angrisani et al.	HMD glasses	☑	Inspection, predictive maintenance	Temperature/humidity sensor, Micro-computer	BCI, EEG, SSVEP, Bluetooth	Bluetooth	Laboratory
Mourtzis et al.	Tablet	☑	Predictive maintenance	-	Cloud, Database, XML, Vuforia, IaaS, HTTPS, RoR, MongoDB-NoSQL	Markers	Laboratory
Verde et al.	HoloLens2	☑	Maintenance, safety	Termocamera Flix	CPE, MEC, 5G network	Depth sensors	Laboratory
Rosales et al.	HoloLens2, iPad	☑	Predictive maintenance	PLC sensors	OPC sever, ThingWorx, ThingGenerator, SCADA, Microsoft Azure	Spatial anchors	Industrial company
Lalik and Watorek	HoloLens2	☑	Predictive maintenance	Camera Photron	NET1-HF, Cloud, Wifi	Markers	Laboratory
Kim et al.	Smartphone	☐	Manufacturing	RGB-D camera	Mask R-CNN, TensorFlow, Pytorch	Camera, deep learning	Laboratory
Husinsky et al.	HoloLens2	☑	Manufacturing, maintenance	Raspberry Pi 4	OPC-UA, MQTT, Node-RED, Okino Nugraf, CAD	3D CAD	Industrial company
Stacchio et al.	HoloLens2	☐	Manufacturing	-	REST API, framework Flask	-	Laboratory
Gross and Wuest	HMD glasses	☐	Manufacturing	Arduino Nano, RFID, OTS	Kivy, Excel	Arduino	Laboratory

4 Results and Discussion

In the study, a comparative analysis of RFID sensor technology and off-the-shelf (OTS) sensors was performed and they were found to underperform, especially in the areas of communication, data processing and customisation capabilities [44]. It is important to test the proposed systems combining data and visualisation in an augmented reality environment with users and obtain feedback. This step is essential to understand the user acceptability of the proposed system. In an experiment with the proposed system, a 10% increase in user awareness of machine status was confirmed [38]. Rosales et al. [40] further supported the results showing that data visualisation reduced troubleshooting time from 50 minutes to 35-40 minutes, an improvement of 20% to 30% depending on the technicians' skills. Visualisation greatly helped users to identify and quickly repair the source of the fault.

In defining the content of the paper for a detailed search of the selected narrow topic, we came across several studies that are closely related to the topic. These publications deal with IoT and visualisation but are not specifically focused on industrial manufacturing environments or engineering applications within Industry 4.0. However, deployment and working with data are relevant across different domains. Alonso-Rosa et al. [45] use AR to visualise energy consumption data in a building environment that is used by maintenance teams. Similar innovative approaches can be found in the area of virtual shopping using AR [46] and in water management [47]. A pure research approach that explores the connection between HMD glasses and a database via an Arduino Uno R3 is explored by Aguero et al. [48]. A key subject that is often associated with AR and IoT communication is the creation of virtual twins, which can in many ways replace AR environments, as VR is more technologically advanced in many aspects [49], [50].

Furthermore, the authors' interest in the area of manufacturing and predictive maintenance is evident, as illustrated by publications that extend this area within IoT in Industry 4.0 [51], [52], [53], [54], [55].

5 Conclusion

Industry 4.0 aims to digitally transform company processes and systems to increase competitiveness. While many efforts are focused on technologies such as IoT, Cloud Computing and AI, less attention is being paid to extended reality technologies. Yet it appears that AR can be groundbreaking, as indicated by the growing research into industrial augmented reality. Smart infrastructure architecture includes physical and software elements, highlighting the relationship between people and technology.

In this study, we have conducted a narrow systematic review of publications that focus on the use of augmented reality and metadata visualisation in the context of Industry 4.0, with an emphasis on the use of IoT elements, which is one of the pillars of the whole of Industry 4.0. The relevance of this review is mainly in its treatment of only current papers published in 2020 and newer. This period reflects all publications that were registered in the Web of Science database between 2020 and the beginning of 2024. In parallel, we can see a decreasing trend in the number of publications in this selected area.

What is the general motivation for research in IoT and AR?
Augmented Reality support is an effective means to minimise losses in industrial processes and improve product quality. Integrating AR with IoT opens new possibilities in data interpretation and linking, facilitating rapid prototyping, and preventing errors in production.

What hardware and combinations of hardware are used?
The positive aspect is that a significant proportion of users are now using AR hardware in the form of HoloLens 2 HMD glasses or other HMD devices. Often these are combined with external hardware, and within the sample studied, this integration is done for example using the Raspberry Pi 4. This practice is often motivated by the specific requirements of a particular project.

What is the software used and what are the most common algorithms used to detect objects in space?
An interesting phenomenon in the sample studied is the fact that each study works with a different set of software. This fact suggests the existence of a diverse range of software products on the market that can be combined with AR technology. From a broader perspective, a close relationship between software and different variants of cloud solutions can be observed.

What are the common areas of real-world applications of IoT and AR?
Applications that integrate IoT with AR are predominantly focused on predictive or traditional maintenance and manufacturing, with a smaller presence in security. This fact raises the question of why the connection between IoT and AR is not finding widespread use in other industries. Clearly, maintenance and manufacturing appear to be the most robust areas for implementing these technologies.

What is the proportion of connections to real-time data in the sample?
Positively, of the 9 research samples examined, 6 are engaged in real-time data analysis. This data is either directly obtained from sensors located on the production machine or integrated into a database system that dynamically manipulates these values.

Communication using standards such as OPC-UA and data conversion using the MQTT format enables easy real-time data manipulation. The development of artificial intelligence and deep learning strengthens the concept of predictive maintenance and enables object detection using neural networks. Object detection in AR environments can be done using traditional methods such as Bluetooth transducers or innovative techniques such as depth sensors. The combination of spatial anchoring and deep learning algorithms enables accurate placement of virtual content in real-world environments.

Research highlights the potential of AR in predictive maintenance and manufacturing, supported by deep learning algorithms and integration with IoT. The use of real-time data and the use of HMD AR glasses are positive developments. It should be pointed out that most of these technologies are still in the laboratory research phase, but some applications have already been successfully deployed in industrial environments.

Following this article, it is worth considering expanding the search filters for articles in the database by incorporating additional keywords. This could enable us to broaden the scope to include articles that might have been overlooked because they did not contain the keywords used in this work. Therefore, we propose the following keywords for potential expansion of the search: Condition monitoring; Smart factories; Digital twins; Sustainable manufacturing; Remote monitoring; Real-time data analytics; Asset management; Cyber-physical systems; Industrial automation; Supply chain management; Quality control; Fault detection; Process optimisation; Big data analytics; and Cloud computing.

Funding

This work was supported by the Internal Science Foundation of the University of West Bohemia under Grant SGS-2023-025 ‘Environmentally sustainable production’.

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